



TERRA AI

**New generation national-scale
autonomous agricultural intelligence
infrastructure for Africa**

Built in China · Deployed in Africa

中国制造 · 服务非洲

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 kitokokennedy13@gmail.com

 **Project ID 267184**

**HICOOL
2026**





The problem

CRISIS OVERVIEW



The problem

309 million undernourished Africans (FAO). Over 27 million food-insecure people in the DRC. Maize yield in the DRC: 800 kg/ha vs. 4–8 t/ha in neighboring countries. 80 million hectares of arable land in the DRC, with less than 10% under cultivation.



Lack of data :
when to irrigate, what quantity of fertilizer to use, when to treat a disease; there is no soil mapping, no crop monitoring, and no historical data.



Lack of personalized agronomic support: Even when certain information is available, many farmers do not have access to a clear interpretation of their data or to agronomic support tailored to their situation.



Lack of digital infrastructure in rural areas: Many African agricultural areas have limited electricity, poor internet access, and weak network coverage.



Governments and agricultural actors lack data on their own agriculture: how many farmers are producing, where they are, what crops exist, and what output to expect. The result: aid is often generic and poorly targeted.



Terra AI Infrastructure

HOW IT WORKS

SENSE

1. Box

Terra Box: soil sensors (EC, moisture, temperature, pH), leaf and air sensors. Solar-powered and autonomous.



CONNECT

2. Network

LoRaWAN Network: RAK7289 gateway, ~15 km range, up to 1,000 Terra Boxes. No SIM card, no existing network required.



INTELLIGENCE

3. Cloud

Terra Cloud: Private LoRaWAN server + database + sensor, laboratory, and weather data correlation.



DECISION

4. SMS

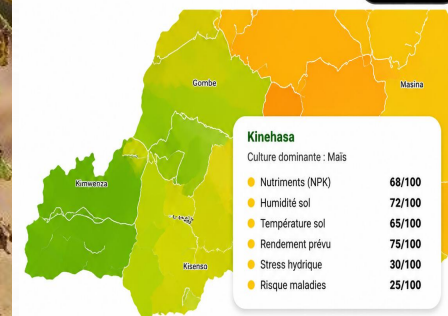
AI → Local-language SMS recommendations on basic mobile phones. Available 24/7



ECOSYSTEM

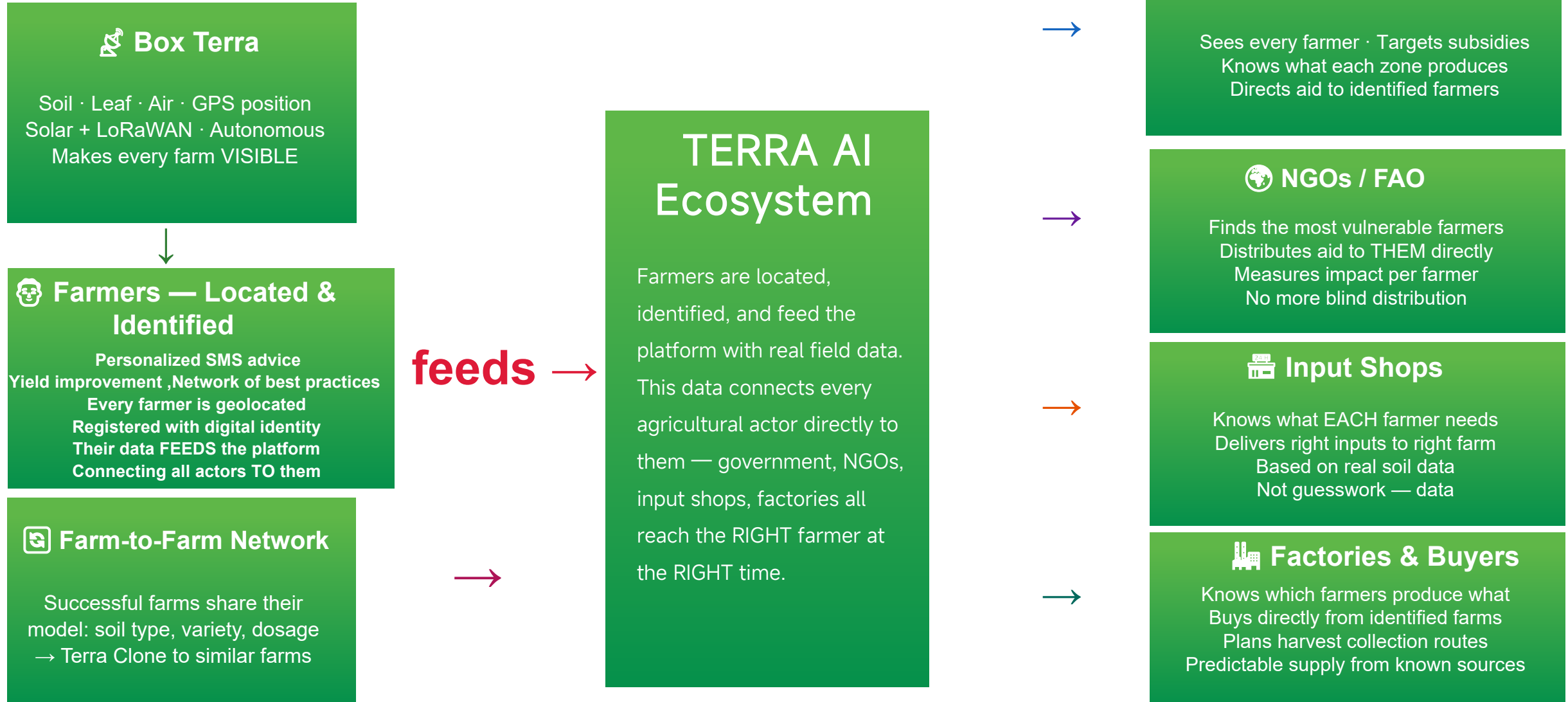
5. PLATFORM

Soil maps, crops, farmers, yields, and risks. This data directly connects all agricultural stakeholders (governments, NGOs, input suppliers, etc.) to farmers.



Terra AI: National Agricultural Platform

One platform connecting every actor in a country's agricultural ecosystem



Terra AI Digital Agricultural Ecosystem

Terra AI provides a complete digital agricultural ecosystem by delivering an autonomous infrastructure adapted to African realities: Terra Box (field sensors), Terra LoRaWAN network, agricultural cloud platforms, and agronomic artificial intelligence.

This ecosystem connects farmers with governments, NGOs, FAO, seed and fertilizer suppliers, researchers, processors, and buyers.



Terra AI Brings Its Own Infrastructure

Everything deployed on-site is ours — energy, network, sensors, intelligence.

100% Made
in China



Box Terra

×5 units · IP67

- Solar panel + battery
- Edge AI (RAK3112)
- Autonomous for months
- Weatherproof enclosure



Soil Sensor

SN-3002 · 5-probe

- Conductivity (EC)
- Moisture & temperature
- pH measurement
- Continuous real-time



Leaf Sensor

SN-3001-YM

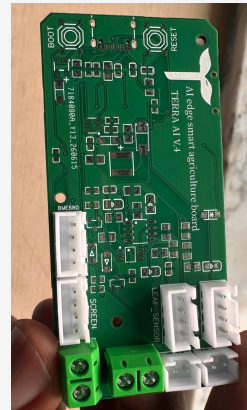
- Leaf temperature
- Leaf wetness
- Disease risk detection
- Crop stress alert



LoRaWAN Gateway

RAK7289 V2

- 16ch LoRaWAN · 15 km
- 1 gateway = 1000 boxes
- Solar powered
- 4G or Satellite uplink



PCB Terra AI V4

×6 boards · JLCPCB

- Our own circuit design
- RAK3112 (ESP32-S3+LoRa)
- All sensor interfaces
- Edge computing ready



Air Sensor

BME688

- Temperature & humidity
- Barometric pressure
- Air quality (VOC)
- Rain detection

FIELD LEVEL: Box ↔ Gateway = LoRaWAN (no SIM, no internet, no telecom) | CLOUD LEVEL: Gateway → Cloud = 4G cellular or Satellite (Starlink) for remote areas

SMS recommendations to farmers require basic mobile network coverage

Complete Field Procedure

FIELD PROCEDURE



Procedure

From Empty Land to Smart Farm

EC mapping → homogeneous zone identification → laboratory analysis per zone → Terra Box installation → crop selection → seasonal monitoring → harvest and continuous improvement.

Step 1-3: Soil Mapping & Laboratory Analysis

Georeferenced EC mapping → detection of homogeneous zones → laboratory analysis by zone (N, P, K, pH, organic matter).



Step 4-5: Terra Box Deployment & Farm Planning

Install Terra Boxes at the most representative points of each zone. Select seeds, fertilizers, application rates, and agricultural schedules.



Step 6-7: Continuous Monitoring & AI Optimization

Continuous monitoring. AI correlates field data with weather conditions. Recommendations are delivered via SMS. Each season enriches the agricultural intelligence database.



Our Scientific Method

SCIENTIFIC METHOD



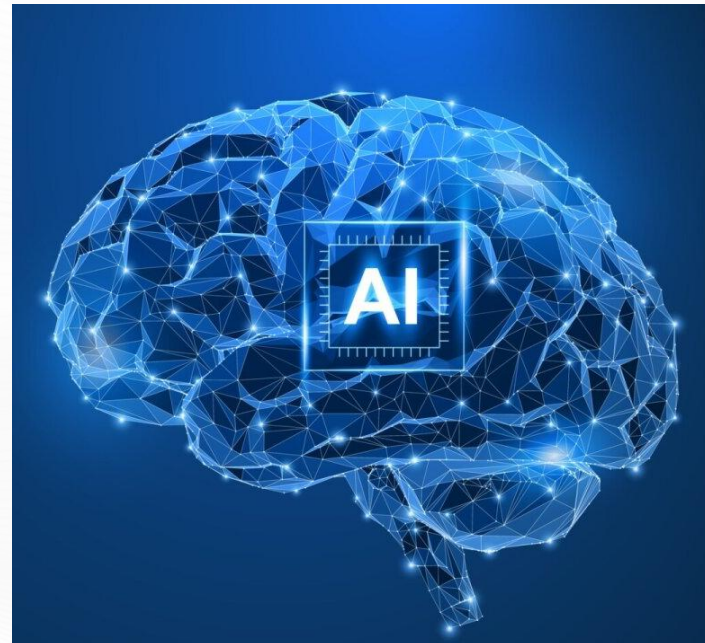
Current Method

The sensor continuously measures EC, moisture, temperature, and pH. NPK values are obtained exclusively through laboratory analysis. The role of the Terra Box is to provide continuous monitoring and alert farmers when a new laboratory analysis is required.



Future Predictive Model

After collecting multiple seasons of data (sensors + laboratory results + real field events), we will train a robust predictive model. Reliable because it is built on real-world field data.



Our Philosophy

We do not skip scientific steps: First, we collect real field data. Then, we build the predictive model — not the other way around.





TERRA AI

北京测试与执行

Beijing Testing & Execution

05:07



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****灌溉建议**:** 暂停灌溉, 待湿度降至30%左右再补水, 避免积水引发病害。

****施肥建议**:** 补充氮磷肥, 建议施用高氮磷、低钾复合肥 (如 20-20-10), 每亩5-8公斤, 沟施后浅覆土。

****风险预警**:**

1. 湿度过高易导致晚疫病, 注意田间排水和通风。
2. 氮磷不足可能影响茎叶生长, 及时追肥。
3. 监测温度变化, 持续高于25℃可能抑制块茎形成。

===== 365s =====

****状态评估**:** 土壤湿度偏高, 温度适宜, pH中性, 钾含量充足, 氮磷偏低。

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===== 384s =====

****状态**:** 土壤湿度偏高, 温度适宜, pH中性, 钾含量充足但氮磷偏低。

****灌溉**:** 暂停灌溉, 待湿度降至30%左右再少量补水。

****施肥**:** 补充氮磷肥, 建议使用高氮磷低钾复合肥 (如 20-20-10), 每亩追施15-20公斤。

****警报**:** 湿度过高易引发根腐病, 注意排水, 避免积水引发根腐病迹象。

===== 405s =====

****状态评估**:** 土壤湿度偏高, 温度适宜, pH中性, 钾含量充足, 氮磷偏低。

****灌溉建议**:** 暂停灌溉, 待湿度降至30%左右再少量补水。



关键洞察 / Key Insight from the Field

“对于没有种植经验的农户来说, 与AI 对话来快速管理土地会有很大帮助。”

“For farmers without experience, talking to AI to quickly manage their land would help greatly.”

— 张洪源 / Zhang Hongyuan, partner farmer Beijing (Feb 2026)

解决方案 / SOLUTION



Competition

COMPETITION



7/7

Terra AI is the only complete solution.



Competition Benchmark

Terra AI: The Only Complete Solution for Smallholder African Farmers

	Terra AI	Sowit	Apollo	CropX	BBOPT
Soil Monitoring	✓	✓	X	✓	X
Offline	✓	X	X	X	X
Automated Irrigation	✓	X	X	X	X
Local Language SMS	✓	X	~	X	~
National Platform	✓	X	X	X	~
Affordable	✓	X	✓	X	✓
Scientific Calibration	✓	X	X	X	X



TAM 7 Mds \$ · SAM 1,2 Md \$ · SOM 60 M \$

PART.03

Market & Traction

Precision Agriculture in Africa. B2G, B2B, and B2C channels. Sales, rental, and yield-sharing business models.



Traction & Validations

TRACTION

China Agricultural University (CAU)

Tested with agronomists trained at the Sino-African Agricultural Science and Technology Courtyard (科技小院).



EquityBCDC Bank

The CEO expressed interest in providing financial support and strategic accompaniment.



Embassy of the DRC

Presented twice and validated by the Ambassador (trained as an agronomist).



HICOOL 2026

Semi-finalist in the Modern Agriculture category. Project ID: 267184

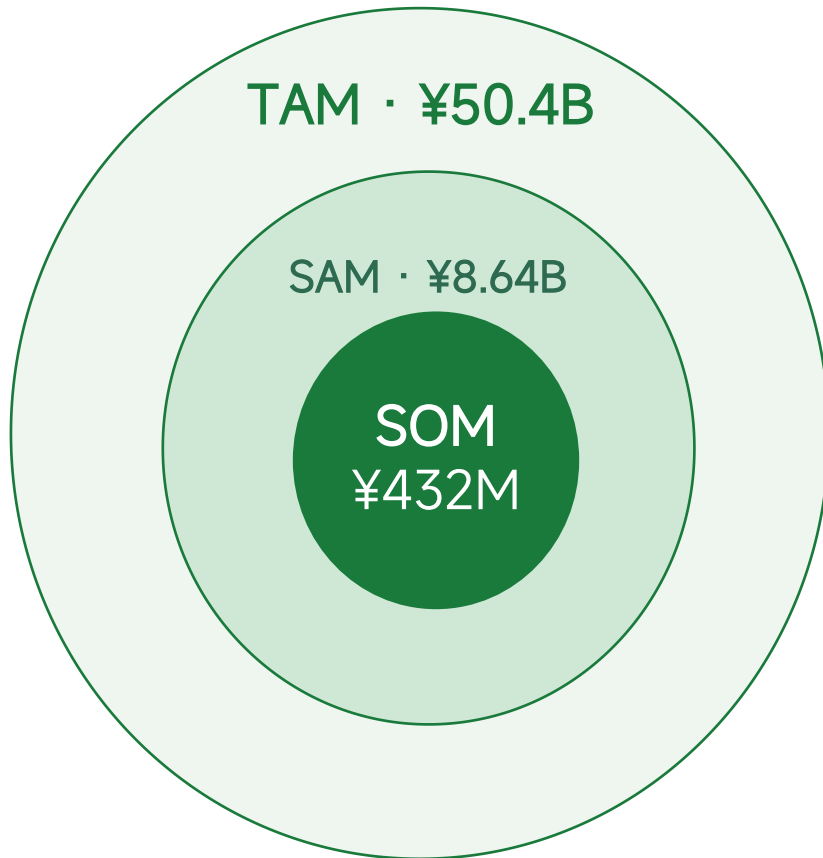


Partner Farmer

Proof of value validated through feedback from the partner farmer.



Market & Access Models



Precision Agriculture in Africa · Sub-Saharan Smallholder Farmers · DRC as the Launchpad

Sales Channels

- 60% B2G — Ministries · World Bank · FAO
- 25% B2B — NGOs · Cooperatives · Data partners
- 15% B2C — Farmer subscriptions (Mobile Money)

Access & Business Models

- Hardware sales + monthly subscription (recurring revenue)
- Seasonal rental model (paid upfront)
- Yield-sharing model (2–5% after harvest)
- One-shot services: soil mapping + laboratory analysis + farm profiling
- Inspired by Soil Health Card (India) and NSMP (Australia)

Why Now

WHY TERRA AI

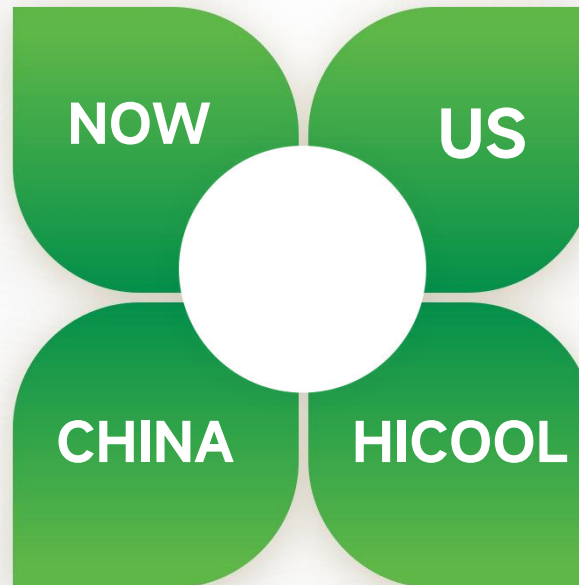
WHY NOW

African food crisis + digital transformation of agriculture
27 million food-insecure people in the DRC
Smart agriculture market growing rapidly
Strong momentum for China–Africa agricultural cooperation.

WHY CHINA

A unique technology and industrial ecosystem
Hardware ecosystem: RAKwireless, JLCPCB
Talent pool: BIT + CAU (985/211 universities)
AI infrastructure: DeepSeek
Belt and Road Initiative
Terra Box cost below \$300

WHY NOW / CHINA / US / HICOOL



WHY US

The right team at the right location
Congolese founder + Beijing Institute of Technology + Embassy support
Multidisciplinary team: hardware, business, agronomy
Field-tested equipment
Terra AI V4 PCB entering production

WHY HICOOL

Beijing as a strategic launch platform

Industrial scaling capability
Investor and government networks
China–Africa validation pathway



TERRA AI

团队介绍



Kennedy Kitoko Muyunga

创始人 · CEO

Founder · CEO

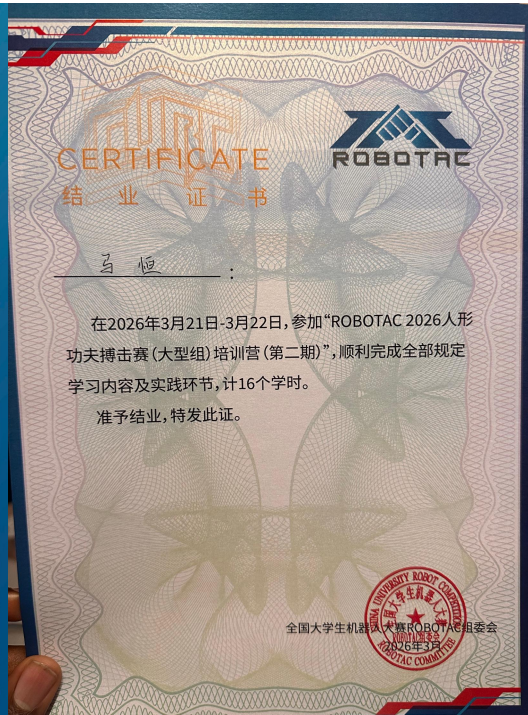


北京理工大学

Beijing Institute of Technology

机电工程 (985 + 211)

Hardware · Edge AI · DRC Embassy
Vice Academic Advisor · ROBOTAC
2026 Champion





团队介绍 *equipe*



Aurélien Tshiamu Tshondo

首席商务官 · CBO

Chief Business Officer

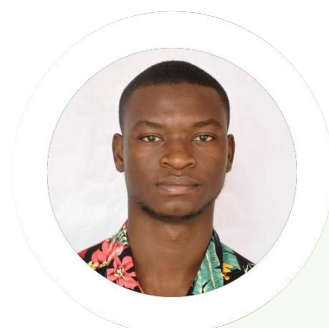


北京理工大学

Beijing Institute of Technology

经济与国际贸易 (985 + 211)

Economics & International Trade



Kibwe Nyandwe Promis

首席农艺师 · CAO

Chief Agronomist



中国农业大学

农艺学硕士 (985 + 211)

China Agricultural University

M.Sc. Agronomy



张乡夫

集成电路与电子学院教授

School of Integrated Circuits and Electronics

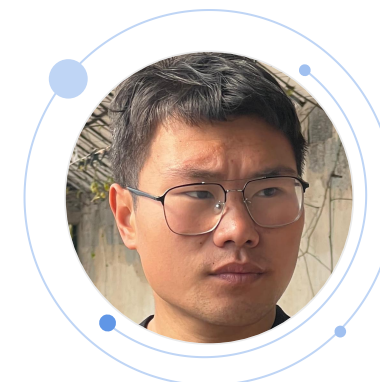


北京理工大学

Beijing Institute of Technology

学术导师 — 硬件验证

Academic Mentor — Hardware Validation



张洪源

悦美农场

farmer

合作农户

Beijing

草莓种植 · 1 个月
验证测试 (2026年
2月)

学术导师 / Academic Mentor: 北京理工大学, 集成电路与电子学院教授

合作伙伴及生态系统

Partners & ecosystem



Beijing Institute of Technology
北京理工大学



China Agricultural University
中国农业大学



JLCPCB



RAKwireless



DRC Embassy · Beijing



HICOOL



Congo bank

BEIJING LANDING PLAN

SLIDE 1/3

Company Registration

JLCPCB → PCB Terra AI V4

RAKwireless → Gateway

RAK7289 + LoRa module

Sensor suppliers → soil, leaf, air

DeepSeek → AI agronomic
engine

Supply Chain & AI

Prof. Zhang Xiangfu (BIT) —
hardware mentor

China Agricultural
University — field validation

IP: trademark + software
copyright + design patent

Beijing: Technology Base

Kitoko AI Technology Co., Ltd.

Pinggu District — Zhongguancun Zone

Kennedy Kitoko = Legal Representative

Social security + income tax paid in Beijing

100%

Made in China

If we leave China (beijing), we
lose the entire supply chain.
Beijing IS the product.

BIT

985+211 University

Beijing: Our Technology Base

DEPLOYMENT PLAN

SLIDE 2/3

① Produce (Q4 2026)

Phase 2: Deploy on 2-4 farms in
DRC
Kinshasa rural
Multiple soil types, multiple
crops
3-6 months field validation

② Test (Q1 2027)

Phase 3: After validation → 100
Boxes
Measure impact: yield, water,
inputs
Sign MoU Ministry of Agriculture
Prepare Series A fundraising

Produce → Test → Scale

Phase 1: Produce 20 test Boxes in Beijing
Validate at BIT lab + IP filings
Ship to DRC for real-world testing

20

Test Boxes first

Contact → Diagnostic → MoU
→ Pilot
→ Impact Proof → Gov.
Contract → Expansion

100

Boxes after
validation

DRC Deployment Plan

BUDGET & MILESTONES

SLIDE 3/3

BEIJING = Tech & Production

DRC Subsidiary (Phase 2)
Installation, training, local support
Government & NGO relationships
Revenue collection (mobile money)

AFRICA = Market & Deploy

- 18-Month Targets:
- 100 Boxes deployed in DRC
 - First revenue generated
 - MoU Ministry of Agriculture
 - Series A preparation

Structure & Budget

Kitoko AI Technology Co., Ltd.

R&D, hardware production, AI development

IP ownership, supplier management

Platform hosting & maintenance

¥72,000 RMB

12-18 months: R&D Beijing + DRC Pilot

Made in Beijing · Deployed in Africa
中国制造 · 服务非洲

60%

B2G + 25% B2B + 15% B2C

Structure, Budget & Milestones

Why Terra AI Will Succeed

The science is proven. The technology makes it scalable.



The Proof: China-Africa STB Program

2,500 kg/ha → 9,000 kg/ha

×3 yield — proven in Malawi with good agronomic science alone, without any technology.

Validated by FAO · Bill & Melinda Gates Foundation · China Agricultural University

But limited to ~30 farms per trained agronomist.

Terra AI: The Same Science, Made Scalable

1 agronomist = 30 farms → 1 Box = 1 farm, 24/7, 365 days

Human-dependent → Technology-powered

Local demonstration → National infrastructure

Knowledge stays with people → Knowledge is in the platform
1 gateway = 1,000 Boxes = 1 entire region covered

DRC: 26M people in food crisis (2026)

80M ha arable land — <10% cultivated

800 kg/ha maize yield

No soil data, no farmer registry

No national agricultural vision



What Terra AI brings — what no other method does

A single infrastructure connecting ALL actors: government, farmers, NGOs, input shops, factories, buyers.

Structured, scalable, data-driven — from the field to the consumer.

Current methods (training, subsidies, aid) are fragmented and not scalable.

Terra AI replaces intuition with data, at national scale.

Terra AI doesn't feed Africa. It gives Africa the infrastructure to feed itself.

From the sensor to the SMS — every data point feeds a decision, every decision feeds a person.



TERRA AI

谢谢

From Field Data to Farmer Action — Turning Every Measurement into Intelligence, and Every Decision into Impact. Made in China · Deployed in Africa.

Kennedy Kitoko Muyunga



kitokokennedy13@gmail.com



Project ID 267184

